

Ansible-02

Tasks :

1. Watch ansible-02 video and write down notes.
2. Install httpd using ansible playbook, use handlers,notifiers.
3. Write a ansible playbook to install apache tomcat.
4. Write a ansible playbook to provision one ec2 on aws.
5. Write a ansible playbook to copy one file from node-1 to node-2.
6. Write a ansible playbook to create different files with different names using single playbook.

Task 2:

2. Install httpd using ansible playbook, use handlers,notifiers.

Step 1:

—> first connect to your two workers servers with Ssh passwordless authentication

—> generate ssh-keygen in ansible controller

—> copy pub-key and paste it inside the worker1(node1)and worker2(node2)—>.ssh—>authentication-keys

—> add paste the public key and save exit

```
root@ip-172-31-2-72:~# ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kg2gYOhRjy2ZkzrKJoFzA5WhP+KC4DMF90fs0lG5PjA root@ip-172-31-2-72
The key's randomart image is:
+--[ED25519 256]--+
  .o  ..
  .o  .  .
  .o  .  +
  o.o  . +E..
  *oo..oo+S
  @o+o .oo o
  oX* . .
  =X+o
  O+=
+----[SHA256]-----+
root@ip-172-31-2-72:~# cat /root/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID5LTgnUy2XDhyWRs5M3VNkekezuSun+bD+qqnUp87xT root@ip-172-31-2-72
root@ip-172-31-2-72:~# ansible all -m ping
```

```

root@ip-172-31-33-34:~# cd .ssh
root@ip-172-31-33-34:~/.ssh# ls
authorized_keys
root@ip-172-31-33-34:~/.ssh# vi authorized_keys
root@ip-172-31-33-34:~/.ssh#

```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 13.60.174.196 PrivateIPs: 172.31.33.34

```

ssh-ed25519 AAAAC3NzaC1lbnR5AAAAD5LTgnUy2XDhyWRs5M3VNkekezuSun+bd+qgnUp87xT root@ip-172-31-2-72
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ubuntu\" rather than the user \"root\".\";echo;sleep 10;exit 142\" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDWNExM5oA0nWRicv1796/gbv0217yuG7Pm0hEG72GLvOWroKziBsWmQCCF02wTlXlRh3R2pIdJv/SdkpbBJNUibPpSE0OHYS+n2/MRki35CS2nVH5usGKmVJM/Ykhh6cyRdPPUVVFNmIADvxshsLpxI3P7kBh9Wk8hVA8P0diwPD3H
puyPxbE3qSN2RXeImHSr1pb4tM+ed8RwR/QDSSpmH213Siyr/+U4u3MXikWAC3GI8GWPm3oew5x3PVW4ktjAxbeB5tIQP8gedeGtfGgq0Lv+sJVV2u6ELUHGeYXAwFTXP2Wchhd/HRjiFLmNQAiQyt+E2GEIugx1TWUafz ansible

```

—> similarly in worker2 also

```

root@ip-172-31-37-159:~# cd .ssh
root@ip-172-31-37-159:~/.ssh# vi authorized_keys
root@ip-172-31-37-159:~/.ssh#

```

i-09212a024d7a75d98 (worker2)

PublicIPs: 13.60.46.139 PrivateIPs: 172.31.37.159

now check in ansible master server

—>ansible all -m ping

```
root@ip-172-31-2-72:~# ansible all -m ping
172.31.37.159 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
172.31.33.34 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
root@ip-172-31-2-72:~# ls
```

Step2:

Create Playbook Folder

—> mkdir playbooks

```
root@ip-172-31-2-72:~# mkdir ~/playbooks
root@ip-172-31-2-72:~# ls
playbooks  snap
root@ip-172-31-2-72:~# cd playbooks/
root@ip-172-31-2-72:~/playbooks# ls
root@ip-172-31-2-72:~/playbooks# vi apache.yml
```

—>cd playbook

Now inside playbooks directory

—> vi apache.yml

Now add below code and save exit

```

---
- hosts: all
  become: yes
  tasks:
    - name: Install Apache
      apt:
        name: apache2
        state: latest
        update_cache: yes

    - name: Copy index.html
      copy:
        src: index.html
        dest: /var/www/html/index.html
      notify: Restart apache2  # <-- aligned with copy, not inside it

  handlers:
    - name: Restart apache2
      service:
        name: apache2
        state: restarted
        enabled: yes

```

```

- hosts: all
  become: yes
  tasks:
    - name: Install Apache
      apt:
        name: apache2
        state: latest
        update_cache: yes

    - name: Copy index.html
      copy:
        src: index.html
        dest: /var/www/html/index.html
      notify: Restart apache2  # <-- aligned with copy, not inside
it

```

handlers:

```

- name: Restart apache2
  service:

```

name: apache2
state: restarted
enabled: yes

Playbook explanation (step by step)

Header

```
- hosts: all  
  become: yes
```

- --- → Start of YAML file
- hosts: all → Run on **all servers** in inventory
- become: yes → Run as **root (sudo)**

Task 1: Install Apache

```
- name: Install Apache  
  apt:  
    name: apache2  
    state: latest  
    update_cache: yes
```

What happens:

- Uses **apt module** (Ubuntu)
- Installs **apache2**
- state: latest → installs or upgrades to latest version
- update_cache: yes → runs **apt update** before install



Idempotent:

If Apache is already installed → nothing changes

Task 2: Copy index.html

```
- name: Copy index.html
  copy:
    src: index.html
    dest: /var/www/html/index.html
    notify: Restart apache2
```

What happens:

- Copies `index.html` from **control node**
- Pastes it on **target server**
- Overwrites only if content is different

notify: Restart apache2

- This tells Ansible:
“If this file **changes**, call the handler”

📌 If file is same → **no restart**

Handler section (VERY IMPORTANT 🔥)

```
handlers:
  - name: Restart apache2
    service:
      name: apache2
      state: restarted
      enabled: yes
```

What is a handler?

- A **special task**
- Runs **only when notified**
- Usually used for **service restart/reload**

What this handler does:

- Restarts Apache service
- Ensures Apache starts on reboot (`enabled: yes`)

How notify + handler works (easy flow)

1. Playbook runs
2. Apache installs
3. `index.html` copied
4. If file changed 🖱️ **notify triggered**
5. Handler runs at **end of play**
6. Apache restarts once

Step 3:

Create index.html File

Vi index.html

```
root@ip-172-31-2-72:~/playbooks# vi index.html
root@ip-172-31-2-72:~/playbooks# vi index.html
root@ip-172-31-2-72:~/playbooks# ansible all -m ser
```

```
root@ip-172-31-2-72:~/playbooks# cat index.html
hello white tiger
root@ip-172-31-2-72:~/playbooks#
```

Step 4:

Step 5: Validate Playbook Before Running

From inside ~/playbooks:

Check YAML syntax

ansible-playbook apache.yml --syntax-check

```
root@ip-172-31-2-72:~/playbooks# ansible-playbook apache.yml --syntax-check
playbook: apache.yml
```

If no syntax errors it will show like above photo

—> if any error it will show red lines reasons

—-> now

ansible-playbook apache.yml —check

Step -5 :

Run playbook for real

—-> ansible-playbook apache.yml

```
root@ip-172-31-2-72:~/playbooks# ansible-playbook apache.yml

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Install Apache] *****
changed: [172.31.37.159]
changed: [172.31.33.34]

TASK [Copy index.html] *****
changed: [172.31.37.159]
changed: [172.31.33.34]

RUNNING HANDLER [Restart apache2] *****
changed: [172.31.37.159]
changed: [172.31.33.34]

PLAY RECAP *****
172.31.33.34      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.37.159   : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```


Step 6:

Verify Installation

—> open your worker1 or worker2 both will have apache2 running inside it

```
root@ip-172-31-33-34:~# systemctl status apache2
Jnit apache2.service could not be found.
root@ip-172-31-33-34:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-29 09:43:50 UTC; 33min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 10062 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 10065 (apache2)
    Tasks: 55 (limit: 1017)
   Memory: 5.8M (peak: 6.1M)
      CPU: 122ms
   CGroup: /system.slice/apache2.service
           └─10065 /usr/sbin/apache2 -k start
             └─10067 /usr/sbin/apache2 -k start
               └─10068 /usr/sbin/apache2 -k start

Jan 29 09:43:50 ip-172-31-33-34 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 29 09:43:50 ip-172-31-33-34 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 16.171.162.0 PrivateIPs: 172.31.33.34

← → ↻ ⚠ Not Secure 16.171.162.0



hello white tiger

—> by using use handlers,notifiers.

—> we are able to restart our service when changes in index.html file

Task 3:

3. Write a ansible playbook to install Apache tomcat.

Step 1:

cd ~/playbooks

. Write Playbook (tomcat.yml)

—> Vi tomcat.yml

```
root@ip-172-31-2-72:~/playbooks# vi tomcat.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook tomcat.yml --syntax-check

playbook: tomcat.yml
```

```
--
- hosts: all
  remote_user: root
  become: yes

  tasks:
    - name: Install Java (required for Tomcat)
      apt:
        name: default-jdk
        state: present
        update_cache: yes

    - name: Ensure /opt/tomcat directory exists
      file:
        path: /opt/tomcat
        state: directory
        mode: '0755'

    - name: Download Apache Tomcat 9.0.115
      get_url:
        url: https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.115/bin/apache-tomcat-9.0.115.tar.gz
        dest: /tmp/apache-tomcat-9.0.115.tar.gz

    - name: Extract Tomcat archive into /opt/tomcat
      unarchive:
        src: /tmp/apache-tomcat-9.0.115.tar.gz
        dest: /opt/tomcat
        remote_src: yes
        creates: /opt/tomcat/apache-tomcat-9.0.115

    - name: Create systemd service file for Tomcat
      copy:
        dest: /etc/systemd/system/tomcat.service
        content: |
          [Unit]
          Description=Apache Tomcat 9
          After=network.target

          [Service]
          Type=forking
          Environment=CATALINA_HOME=/opt/tomcat/apache-tomcat-9.0.115
```

```

creates: /opt/tomcat/apache-tomcat-9.0.115
- name: Create systemd service file for Tomcat
  copy:
    dest: /etc/systemd/system/tomcat.service
    content: |
      [Unit]
      Description=Apache Tomcat 9
      After=network.target

      [Service]
      Type=forking
      Environment=CATALINA_HOME=/opt/tomcat/apache-tomcat-9.0.115
      Environment=CATALINA_BASE=/opt/tomcat/apache-tomcat-9.0.115
      ExecStart=/opt/tomcat/apache-tomcat-9.0.115/bin/startup.sh
      ExecStop=/opt/tomcat/apache-tomcat-9.0.115/bin/shutdown.sh
      User=root
      Group=root
      Restart=always

      [Install]
      WantedBy=multi-user.target

- name: Reload systemd daemon
  command: systemctl daemon-reload

- name: Enable Tomcat service
  systemd:
    name: tomcat
    enabled: yes

- name: Start Tomcat service
  systemd:
    name: tomcat
    state: started

```

- hosts: all
 remote_user: root
 become: yes

tasks:

- name: Install Java (required for Tomcat)
 apt:
 name: default-jdk
 state: present
 update_cache: yes

- name: Ensure /opt/tomcat directory exists
 file:
 path: /opt/tomcat
 state: directory
 mode: '0755'

- name: Download Apache Tomcat 9.0.115
get_url:
url: <https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.115/bin/apache-tomcat-9.0.115.tar.gz>
dest: /tmp/apache-tomcat-9.0.115.tar.gz
- name: Extract Tomcat archive into /opt/tomcat
unarchive:
src: /tmp/apache-tomcat-9.0.115.tar.gz
dest: /opt/tomcat
remote_src: yes
creates: /opt/tomcat/apache-tomcat-9.0.115
- name: Create systemd service file for Tomcat
copy:
dest: /etc/systemd/system/tomcat.service
content: |
[Unit]
Description=Apache Tomcat 9
After=network.target

[Service]
Type=forking
Environment=CATALINA_HOME=/opt/tomcat/apache-tomcat-9.0.115
Environment=CATALINA_BASE=/opt/tomcat/apache-tomcat-9.0.115
ExecStart=/opt/tomcat/apache-tomcat-9.0.115/bin/startup.sh
ExecStop=/opt/tomcat/apache-tomcat-9.0.115/bin/shutdown.sh
User=root
Group=root
Restart=always

[Install]

WantedBy=multi-user.target

- name: Reload systemd daemon
command: systemctl daemon-reload

- name: Enable Tomcat service
systemd:
name: tomcat
enabled: yes

- name: Start Tomcat service
systemd:
name: tomcat
state: started

--> save and exit :wq!

Step 2:

--> ansible-playbook tomcat.yml --syntax-check

Now apply

--> ansible-playbook tomcat.yml

```

root@ip-172-31-2-72:~/playbooks# ansible-playbook tomcat.yml

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [172.31.33.34]
ok: [172.31.37.159]

TASK [Install Java (required for Tomcat)] *****
changed: [172.31.37.159]
changed: [172.31.33.34]

TASK [Ensure /opt/tomcat directory exists] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Download Apache Tomcat 9.0.115] *****
changed: [172.31.33.34]
changed: [172.31.37.159]

TASK [Extract Tomcat archive into /opt/tomcat] *****
changed: [172.31.37.159]
changed: [172.31.33.34]

TASK [Create systemd service file for Tomcat] *****
changed: [172.31.33.34]
changed: [172.31.37.159]

TASK [Reload systemd daemon] *****
changed: [172.31.37.159]
changed: [172.31.33.34]

TASK [Enable Tomcat service] *****
changed: [172.31.33.34]
changed: [172.31.37.159]

TASK [Start Tomcat service] *****
changed: [172.31.33.34]
changed: [172.31.37.159]

PLAY RECAP *****
172.31.33.34      : ok=9    changed=7    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.37.159    : ok=9    changed=7    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Key Modules Used

- apt → Installs Java
- file → Creates /opt/tomcat directory
- get_url → Downloads Tomcat archive
- unarchive → Extracts Tomcat into /opt/tomcat
- copy → Creates tomcat.service systemd file
- command → Reloads systemd daemon
- systemd → Enables and starts Tomcat

Verify Tomcat service

Step 3:

Verify in workers servers

—> check if tomcat running or not

```
root@ip-172-31-33-34:/opt# systemctl status tomcat
● tomcat.service - Apache Tomcat 9
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-29 10:46:14 UTC; 22s ago
     Process: 14639 ExecStart=/opt/tomcat/apache-tomcat-9.0.115/bin/startup.sh (code=exited, status=0/SUCCESS)
    Main PID: 14649 (java)
      Tasks: 29 (limit: 1017)
     Memory: 86.1M (peak: 86.6M)
        CPU: 3.585s
    CGroup: /system.slice/tomcat.service
            └─14649 /usr/bin/java -Djava.util.logging.config.file=/opt/tomcat/apache-tomcat-9.0.115/conf/logc

Jan 29 10:46:14 ip-172-31-33-34 systemd[1]: Starting tomcat.service - Apache Tomcat 9...
Jan 29 10:46:14 ip-172-31-33-34 startup.sh[14639]: Tomcat started.
Jan 29 10:46:14 ip-172-31-33-34 systemd[1]: Started tomcat.service - Apache Tomcat 9.
lines 1-14/14 (END)
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 16.171.162.0 PrivateIPs: 172.31.33.34

← → 🔍 Not Secure 16.171.162.0:8080

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.115

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

Tomcat Setup	Realms & AAA	Examples	Servlet Specifications
First Web Application	JDBC DataSources		Tomcat Versions

Task 4:

4. Write an ansible playbook to provision one ec2 on aws.

Prerequisites

Install AWS libraries for Ansible

```
sudo apt install -y python3-boto3 python3-botocore
```

Install AWS CLI (Ubuntu 24.04)

```
sudo snap install aws-cli --classic
```

```
root@ip-172-31-2-72:~/playbooks# apt install -y python3-boto3 python3-botocore
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-boto3 is already the newest version (1.34.46+dfsg-1ubuntu1).
python3-boto3 set to manually installed.
python3-botocore is already the newest version (1.34.46+repack-1ubuntu1).
python3-botocore set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 30 not upgraded.
root@ip-172-31-2-72:~/playbooks# sudo apt update
```

```
root@ip-172-31-2-72:~/playbooks# sudo snap install aws-cli --classic
aws-cli (v2/stable) 2.33.9 from Amazon Web Services (aws✓) installed
```


Step 1:

Configure AWS credentials

aws configure

```
root@ip-172-31-2-72:~/playbooks# aws configure
AWS Access Key ID [None]: AKIATAWNKXROY7TFMYF
AWS Secret Access Key [None]: fg1I55ElLDWVXT3qTQbEl0yp3+uAgu9XpUN+Z05I
Default region name [None]: us-east-1
Default output format [None]: json
root@ip-172-31-2-72:~/playbooks#
```

Step 2:

Write your playbook (ec2.yml)

Inside your play book create ec2.yml → vi ec2.yml and write down below code

```
gather_facts: false

collections:
  - amazon.aws

tasks:
  - name: Launch EC2 instance
    amazon.aws.ec2_instance:
      name: "ansible-demo-instance"
      key_name: "ansible" # replace with your AWS key pair name
      instance_type: t3.micro
      image_id: ami-04233b5aecce09244
      region: eu-north-1
      wait: true
      count: 1
      security_group: default
      network:
        assign_public_ip: true
      register: ec2

  - name: Show instance info
    debug:
      var: ec2.instances
```

"ec2.yml" 28L, 636B

```
root@ip-172-31-2-72:~/playbooks# vi ec2.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook ec2.yml --syntax-check
playbook: ec2.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook ec2.yml
```

--> ansible-playbook ec2.yml --syntax-check
To check if the code is correct

--> now apply
--> ansible-playbook ec2.yml

```
root@ip-172-31-2-72:~/playbooks# ansible-playbook ec2.yml
PLAY [Provision EC2 instance in AWS] *****
TASK [Launch EC2 instance] *****
changed: [localhost]
TASK [Show instance info] *****
ok: [localhost] => {
  "ec2.instances": [
    {
      "ami_launch_index": 0,
      "architecture": "x86_64",
      "block_device_mappings": [
        {
          "device_name": "/dev/xvda",
          "ebs": {
            "attach_time": "2026-01-29T11:36:57+00:00",
            "delete_on_termination": true,
            "status": "attaching",
            "volume_id": "vol-0feee9ec28e66e8b6"
          }
        }
      ],
      "boot_mode": "uefi-preferred",
      "capacity_reservation_specification": {
        "capacity_reservation_preference": "open"
      },
      "client_token": "9676509ab1194cfba1660535074de2e1",
      "cpu_options": {
        "core_count": 1,
        "threads_per_core": 2
      },
      "current_instance_boot_mode": "uefi",
      "ebs_optimized": false,
      "ena_support": true,
      "enclave_options": {
        "enabled": false
      },
      "hibernation_options": {
        "configured": false
      },
      "hypervisor": "xen",
      "image_id": "ami-04233b5aecce09244",
      "instance_id": "i-0797843b9097bb3a6",
      "instance_type": "t3.micro",
      "key_name": "ansible",
    }
  ]
}
```

```

    },
    "source_dest_check": true,
    "state": {
      "code": 0,
      "name": "pending"
    },
    "state_transition_reason": "",
    "subnet_id": "subnet-0a094896b5cb2db9e",
    "tags": {
      "Name": "ansible-demo-instance"
    },
    "usage_operation": "RunInstances",
    "usage_operation_update_time": "2026-01-29T11:36:56+00:00",
    "virtualization_type": "hvm",
    "vpc_id": "vpc-01126d967b1bb73af"
  }
}
}

PLAY RECAP *****
localhost                : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@ip-172-31-2-72:~/playbooks# vi ec2.yml
root@ip-172-31-2-72:~/playbooks#

```

- > now check your instances created or not
- > go to instances

☐	worker1	i-0e57dab8fabc0e480	Running	t3.micro	3/3 checks passed	View alarms	eu-north-1b	ec2-16-171-162-0.eu-n...	16.171.162.0
☐	worker2	i-09212a024d7a75d98	Running	t3.micro	3/3 checks passed	View alarms	eu-north-1b	ec2-16-171-31-3.eu-no...	16.171.31.3
☒	ansible-demo-...	i-0797843b9097bb3a6	Running	t3.micro	Initializing	View alarms	eu-north-1a	ec2-13-51-106-212.eu-...	13.51.106.21

Task 5:

5. Write an Ansible playbook to copy one file from node-1 to node-2.

By default, Ansible's `copy` module copies from the controller → node, not node → node.

To move files between nodes, you need to:

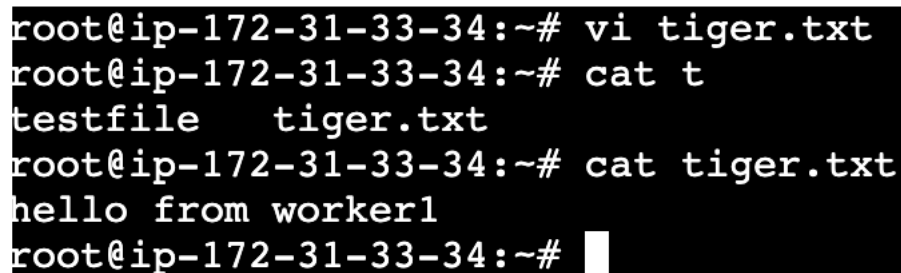
1. **Fetch the file** from node-1 to the controller.
2. **Copy the file** from the controller to node-2.

Step1:

Create a file in worker1 (node-1)

—> vi tiger.txt

—> text inside file is hello from worker1

A terminal window with a black background and white text. The prompt is root@ip-172-31-33-34:~#. The user enters 'vi tiger.txt', then 'cat t', which results in an error 'testfile tiger.txt'. Then the user enters 'cat tiger.txt' and the output is 'hello from worker1'.

```
root@ip-172-31-33-34:~# vi tiger.txt
root@ip-172-31-33-34:~# cat t
testfile  tiger.txt
root@ip-172-31-33-34:~# cat tiger.txt
hello from worker1
root@ip-172-31-33-34:~#
```

—> now go to ansible controller server

Step -2 create copy.yml in playbook

```
root@ip-172-31-2-72:~/playbooks# vi copy.yml
root@ip-172-31-2-72:~/playbooks#
```

```
tasks:
  - name: Fetch file from worker1 to control node
    fetch:
      src: /root/tiger.txt
      dest: /tmp/tiger.txt
      flat: yes

  - name: Copy tiger.txt to worker2
    hosts: worker2
    become: yes
    tasks:
      - name: Copy file from control node to worker2
        copy:
          src: /tmp/tiger.txt
          dest: /root/tiger.txt
```

```
"copy.yml" 20L, 419B
```

hosts: worker1

become: yes

tasks:

- name: Fetch file from worker1 to control node

fetch:

src: /root/tiger.txt

dest: /tmp/tiger.txt

flat: yes

- name: Copy tiger.txt to worker2

hosts: worker2

become: yes

tasks:

- name: Copy file from control node to worker2

copy:

src: /tmp/tiger.txt

dest: /root/tiger.txt

Step 3:

—>now first check the syntax

—-> ansible-playbook copy.xml —syntax-check

—-> ansible-playbook copy.xml

```
root@ip-172-31-2-72:~/playbooks# ansible-playbook copy.yml --syntax-check
playbook: copy.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook copy.yml

PLAY [Fetch tiger.txt from worker1] *****
TASK [Gathering Facts] *****
ok: [172.31.33.34]

TASK [Fetch file from worker1 to control node] *****
changed: [172.31.33.34]

PLAY [Copy tiger.txt to worker2] *****
TASK [Gathering Facts] *****
ok: [172.31.37.159]

TASK [Copy file from control node to worker2] *****
changed: [172.31.37.159]

PLAY RECAP *****
172.31.33.34      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.37.159   : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@ip-172-31-2-72:~/playbooks#
```

— ->Check the file in wokernode -02

```
root@ip-172-31-33-34:~# ls
ansible.log  snap  testfile  tiger.txt
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 16.171.162.0 PrivateIPs: 172.31.33.34

```
root@ip-172-31-37-159:~# ls
ansible.log  snap  testfile  tiger.txt
root@ip-172-31-37-159:~# cat tiger.txt
hello from worker1
root@ip-172-31-37-159:~#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 16.171.31.3 PrivateIPs: 172.31.37.159

Task 6:

6. Write an Ansible playbook to create different files with different names using a single playbook.

Create a file in the playbook name `multiplefiles.yml`

—> `vi multiplefile.yml`

```
root@ip-172-31-2-72:~/playbooks# vi multiplefiles.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook multiplefiles.yml --syntax-check
```

```
tasks:
  - name: Create different files in /root
    file:
      path: "/root/{{ item }}"
      state: touch
    loop:
      - tigers.txt
      - lion.txt
      - elephant.txt
      - monkey.txt
```

—> add the above code in `multiplefiles.yml`

—> save

—> now check the syntax-check

—> ansible-playbook multiplefiles.yml --syntax-check

When the check is perfect

Run

—> ansible-playbook multiplefiles.yml

```
root@ip-172-31-2-72:~/playbooks# ansible-playbook multiplefiles.yml --syntax-check
playbook: multiplefiles.yml
root@ip-172-31-2-72:~/playbooks# ansible-playbook multiplefiles.yml

PLAY [Create multiple files using single playbook] *****

TASK [Gathering Facts] *****
ok: [172.31.37.159]
ok: [172.31.33.34]

TASK [Create different files in /root] *****
changed: [172.31.37.159] => (item=tigers.txt)
changed: [172.31.33.34] => (item=tigers.txt)
changed: [172.31.37.159] => (item=lion.txt)
changed: [172.31.33.34] => (item=lion.txt)
changed: [172.31.37.159] => (item=elephant.txt)
changed: [172.31.33.34] => (item=elephant.txt)
changed: [172.31.37.159] => (item=monkey.txt)
changed: [172.31.33.34] => (item=monkey.txt)

PLAY RECAP *****
172.31.33.34      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.37.159   : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@ip-172-31-2-72:~/playbooks# vi multiplefiles.yml
```

—> now check any workers server root/

Ls

```
root@ip-172-31-33-34:~# ls
ansible.log snap testfile tiger.txt
root@ip-172-31-33-34:~# ls
ansible.log elephant.txt lion.txt monkey.txt snap testfile tiger.txt tigers.txt
root@ip-172-31-33-34:~#
```

i-0e57dafb81ac0e460 (worker1)

PublicIPs: 16.171.162.0 PrivateIPs: 172.31.33.34

```
root@ip-172-31-37-159:~# ls
ansible.log snap testfile tiger.txt
root@ip-172-31-37-159:~# cat tiger.txt
hello from worker1
root@ip-172-31-37-159:~# ls
ansible.log elephant.txt lion.txt monkey.txt snap testfile tiger.txt tigers.txt
root@ip-172-31-37-159:~#
```

i-09212a024d7a75d98 (worker2)

PublicIPs: 16.171.31.3 PrivateIPs: 172.31.37.159

